

Clearance of intracranial debris by ultrasound reduces inflammation and improves outcomes in hemorrhagic stroke models

Received: 18 September 2024

Accepted: 15 September 2025

Published online: 10 November 2025

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Impaired clearance of neurotoxic debris in the brain exacerbates neurologic disease and presents a promising therapeutic target. Pharmacologic therapies can enhance meningeal lymphatic clearance in preclinical models but may be limited by systemic toxicities or invasive administration. Here we report a low-intensity, focused ultrasound protocol that noninvasively clears pathogenic substances from the cerebrospinal fluid and brain interstitium in mice. Using two models of hemorrhagic stroke, we demonstrate that this protocol clears the cerebrospinal fluid and interstitium of blood cells, which accumulate in the deep cervical lymph nodes via meningeal lymphatics. The protocol directly modulates molecular processes, including mechanosensitive channels, to shift microglial phenotypes and astrocytic aquaporin localization to reduce neuroinflammation and neurocytotoxicity. In the intracerebral hemorrhage model, it improves behavioral outcomes and increases survival with greater efficacy than a pharmacologic benchmark. The protocol satisfies Food and Drug Administration safety guidelines, supporting clinical translatability. If demonstrated effective clinically, it may provide therapeutic benefit not only in hemorrhagic stroke but also in other neurologic disorders that involve impaired debris clearance.

Impairment of cerebrospinal fluid (CSF) circulation has been associated with pathogenesis in various neurologic disorders^{1–3}. In particular, impaired clearance of dispersed debris in CSF and interstitial fluid promotes neuroinflammation and neurocytotoxicity, contributing to pathology and symptom severity in ischemic stroke, hemorrhagic stroke, traumatic brain injury, neurodegenerative diseases and other neurologic conditions^{1–3}. Pharmacologic upregulation of meningeal lymphatics to enhance clearance of neurotoxic debris has demonstrated therapeutic potential in preclinical models^{4–6}. However, despite promising preclinical data, currently there are no

approved pharmacologic therapies that directly enhance intracranial fluid clearance. Concerns about experimental drug candidates relate to off-target effects, systemic toxicities, prolonged latency of therapeutic effects and limited translational feasibility of invasive administration protocols^{7–9}. Meanwhile, surgical approaches, such as direct CSF drainage in subarachnoid hemorrhage (SAH)¹⁰ and hematoma evacuation in intracerebral hemorrhage (ICH)¹¹, provide clinical benefit but are reserved for the most severely affected individuals, are invasive and do not directly correct the underlying molecular and cellular pathophysiology.

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Focused ultrasound (FUS) has been shown to either enhance or suppress neuroinflammatory profiles in the brain, with outcomes largely dependent on sonication parameters and further influenced by coadministration of exogenous agents such as microbubbles¹². Low-intensity FUS paired with microbubbles may disrupt the blood–brain barrier, which clears Alzheimer’s-associated plaques in preclinical models¹³ but also has sterile neuroinflammatory effects¹⁴ and raises safety concerns, given the risk of microbubble inertial cavitation as well as practical concerns in coordinating sonication while the microbubbles rapidly clear with a half-life of minutes¹². Recently, a FUS protocol without microbubbles recapitulated similar behavioral effects in the same Alzheimer’s model, but clearance of plaque and neurotoxic substances was not shown, and the required ultrasound frequency presents clinical challenges for transcranial transmission¹⁵.

Low-intensity FUS is known to activate mechanosensitive ion channels, such as Piezo1, both *in vitro* and in animals *in vivo*^{16,17}. Additionally, pharmacologic agonists of mechanosensitive channels in animal studies have shown promise for neurologic therapeutics development as they can dampen neuroinflammatory states^{18–20} and enhance glymphatic mediators, including meningeal lymphatic vessel sprouting^{21,22}. However, in many cases these agents must be administered invasively, either intraventricularly or intracisternally, or over multiple weeks²¹, making for practical translational challenges. They may also carry risks of systemic toxicity^{23,24}. None are currently approved for human administration. These findings suggest that if ultrasound protocols could be tailored to activate mechanosensitive ion channels, they could promote similar beneficial effects on neuroinflammation and glymphatic circulation while avoiding invasive administration and systemic toxicity risks.

We and others previously demonstrated that low-intensity FUS alone, without coadministration of exogenous agents, enhances CSF circulation^{25–27}. Ultrasound improved the brain distribution of intrathecally administered drugs and facilitated the circulation of cisternally and intraparenchymally administered tracers^{25–27}. Although these studies did not establish therapeutic efficacy in a neurological disease model, they indicate the potential of low-intensity FUS to clear pathogenic substances from the central nervous system and to achieve corresponding therapeutic effects.

In this study, we developed and rigorously validated a low-intensity transcranial FUS protocol aimed specifically at enhancing clearance of neurotoxic debris. We assessed the protocol’s safety and efficacy in two mouse models of hemorrhagic brain injury: SAH and ICH. We examined clearance of micron-scale hemorrhagic debris from both the CSF and interstitial compartments, changes in neuroinflammation and neurocytotoxicity and the underlying cellular and molecular mechanisms of the clearance effect. Our results show that ultrasound treatment enhances neurotoxic debris clearance, reduces neuroinflammation and neurocytotoxicity, improves functional behavioral outcomes and increases survival, with greater safety and efficacy than a pharmacologic benchmark. At the molecular level, we document dependence on mechanosensitive ion channel activation, with an ultrasound-induced shift in microglia to a homeostatic phenotype and with increased polarization of aquaporins in astrocytic vascular endfeet. If validated in clinical trials, this approach would offer a clinically feasible, nonpharmacologic, noninvasive therapeutic strategy for stroke and potentially other neurologic disorders that involve impaired clearance of neurotoxic debris.

Results

Ultrasonic debris clearance enhances intracranial hemorrhage clearance

To assess the utility of low-intensity transcranial FUS to clear neurotoxic debris from the CSF and interstitial brain compartments in stroke models, we first adapted to mice our prior protocol²⁵ that used continuous 650-kHz ultrasound in rats, given the multitude of disease models developed for mice. Although our earlier protocol was effective, its

ultrasound center frequency of 650 kHz presents challenges for translation to larger animals and humans given both the more prominent absorption by the skull (and therefore risk of unsafe heating) as well as the aberrating effect of the skull and human hair compared to lower frequencies like 200–250 kHz (refs. 28,29). We first confirmed that our prior finding of increased CSF–interstitial fluid mixing with ultrasound showed no appreciable reduction when the center frequency was decreased to 250 kHz (Extended Data Fig. 1a,b). Next, because at 250 kHz the ultrasound focus using this $f=1.0$ focused transducer would affect much of the mouse brain at once (Fig. 1a,b), to prevent heat accumulation during application, we used a pulsed protocol with a duty cycle (25%) that we estimated would not yield substantial parenchymal heating based on the bioheat equation³⁰ and simulation and experimental verification (Extended Data Fig. 2), while keeping consistent the duty cycle of ultrasound application at points of the brain that we used in our prior protocol²⁵. Finally, we completed a parameter variation of the applied ultrasound pressure (Extended Data Fig. 1c) to yield a target ultrasound protocol for use in mice. These results showing that ultrasound increases brain parenchymal entry of CSF-dispersed tracers confirm our prior²⁵ findings that low-intensity FUS increases CSF and interstitial fluid exchange.

We next applied this ultrasonic debris clearance (UDC) protocol to two models of hemorrhagic brain injury in which impaired clearance of hemorrhagic products is thought to be pathogenic: SAH and ICH (also known as intraparenchymal hemorrhage)^{31,32}. Additionally, the micron-scale hemorrhagic debris in these models provide an ambitious target for clearance, which, if successful, would suggest the potential to clear a wide variety of neurotoxic debris. In a mouse SAH model (Fig. 1c), UDC treatment reduced the number of red blood cells detected in the CSF compared to sham treatment (mean $\pm$ s.d.; SAH + sham: 338.33 ± 65.55 versus SAH + UDC: 140 ± 44.27 , $P \leq 0.01$; no extravascular red blood cells visible in healthy controls; Fig. 1d). This reduction was associated with a greater accumulation of red blood cells in the deep cervical lymph nodes, indicating efficient clearance through the meningeal lymphatic system (mean $\pm$ s.d.; SAH + sham: 12 ± 4.47 versus SAH + UDC: 24.83 ± 6.85 , $P \leq 0.01$; no extravascular red blood cells visible in healthy controls; Fig. 1d). Similarly, in the ICH model, UDC treatment reduced the number of interstitial red blood cells compared to sham (mean $\pm$ s.d.; ICH + sham: 26.17 ± 7.47 versus ICH + UDC: 11.67 ± 5.32 , $P \leq 0.01$; no extravascular red blood cells visible in healthy controls; Fig. 1e). This clearance was again accompanied by increased red blood cell accumulation in the deep cervical lymph nodes (mean $\pm$ s.d.; ICH + sham: 7.67 ± 4.37 versus ICH + UDC: 14.5 ± 4.32 , $P \leq 0.05$; no extravascular red blood cells visible in healthy controls; Fig. 1e). These findings demonstrate that UDC facilitates the clearance of micron-scale hemorrhagic debris from both the CSF and interstitial compartments to the deep cervical lymph nodes, likely through the meningeal lymphatics, arachnoid cuff exit points³³ or arterial–venous leptomeningeal overlaps³⁴ that are identified as main egress paths for CSF- and interstitial fluid-dispersed solutes.

UDC reduces posthemorrhagic neuroinflammation and neurocytotoxicity

Neuroinflammation and neurocytotoxicity are critical factors that contribute to poor outcomes after hemorrhagic brain injury³⁵. To evaluate the impact of UDC on these processes, we performed immunohistochemical analyses of brain tissue from UDC-treated SAH and ICH models (Fig. 2a,e). UDC treatment reduced the mean fluorescence intensity of IBA1⁺ cells (a commonly used marker of microglia, including both resting and activated states), suggesting an attenuation of microglial responses in both models (mean $\pm$ s.d.; healthy control: 18.5 ± 3.73 ; SAH + sham: 35 ± 8.27 versus SAH + UDC: 24 ± 7.62 , ICH + sham: 44.7 ± 8.09 versus ICH + UDC: 30.5 ± 7.64 , $P \leq 0.05$; Fig. 2b,f). Additionally, astrocytic activation assessed by glial fibrillary acidic protein (GFAP) staining (an intermediate filament protein expressed by

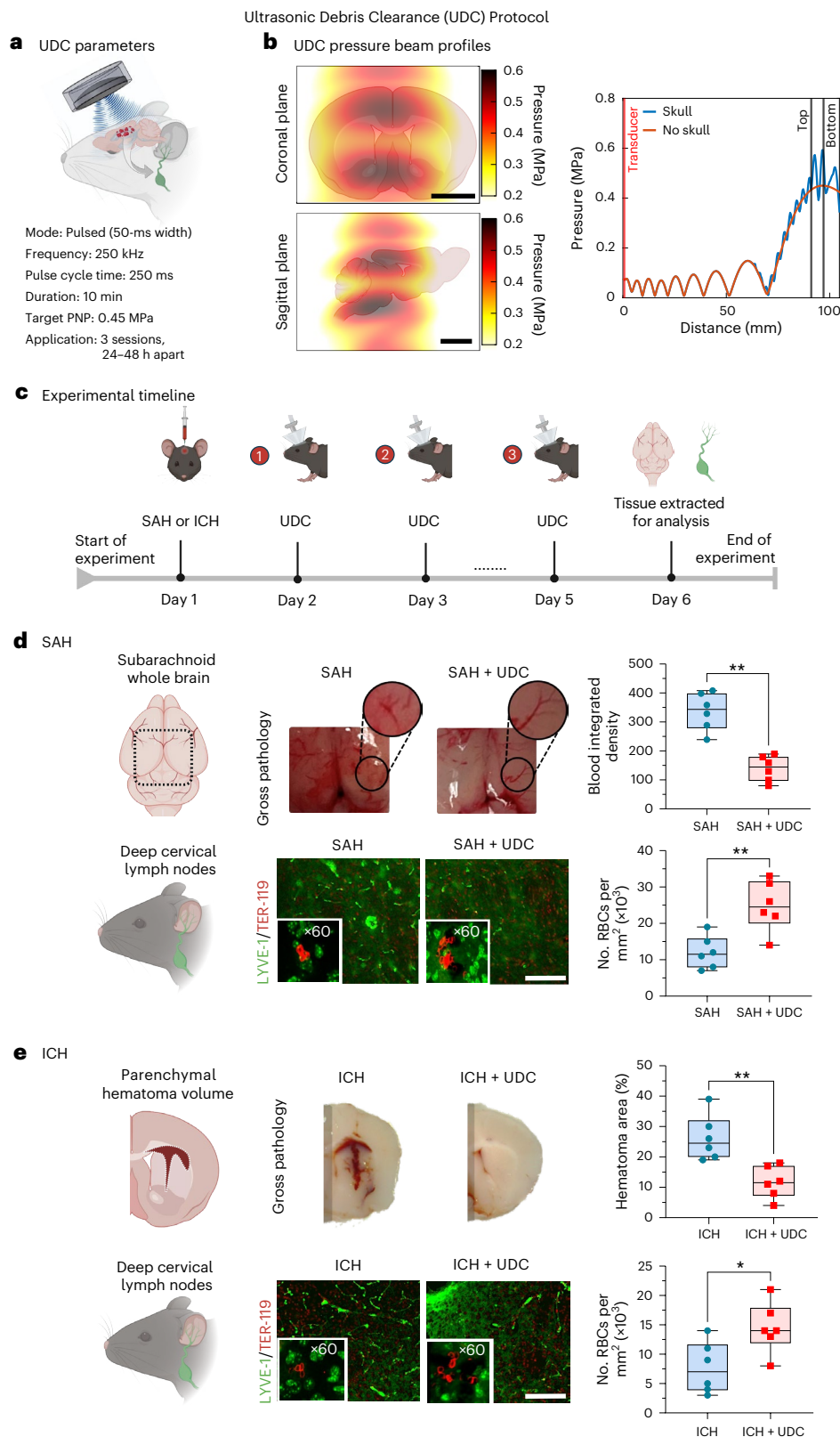


Fig. 1 | UDC enhances the clearance of neurotoxic hemorrhagic debris from the CSF and interstitial brain compartments. **a**, Schematic of the FUS protocol (250 kHz, 0.45-MPa peak negative pressure (PNP), 25% duty cycle, 50-ms pulse width) efflux experiment. **b**, Simulated ultrasound pressure beam profiles and effect of the skull on the intracranial profile. **c**, Timeline of SAH and ICH experiments. **d**, UDC clears CSF-dispersed red blood cells (RBCs) compared to sham in an SAH model ($P = 0.0022$). There is a greater accumulation of red blood cells in the deep cervical lymph nodes of SAH mice with UDC than in those of

SAH sham mice ($P = 0.0087$). **e**, UDC clears interstitial red blood cells compared to sham in an ICH model ($P = 0.0022$). There is a greater accumulation of red blood cells in the deep cervical lymph nodes of ICH mice with UDC than in those with sham ($P = 0.0411$). No red blood cells were observed in separate cohorts of healthy control brain or lymph node tissue. Data presented as box plots that show minima, maxima, interquartile range (box bounds) and median (black line). Data were analyzed by two-tailed Mann–Whitney U -test; * $P \leq 0.05$; ** $P \leq 0.01$; $n = 6$ for each group. Scale bars, 100 μ m.

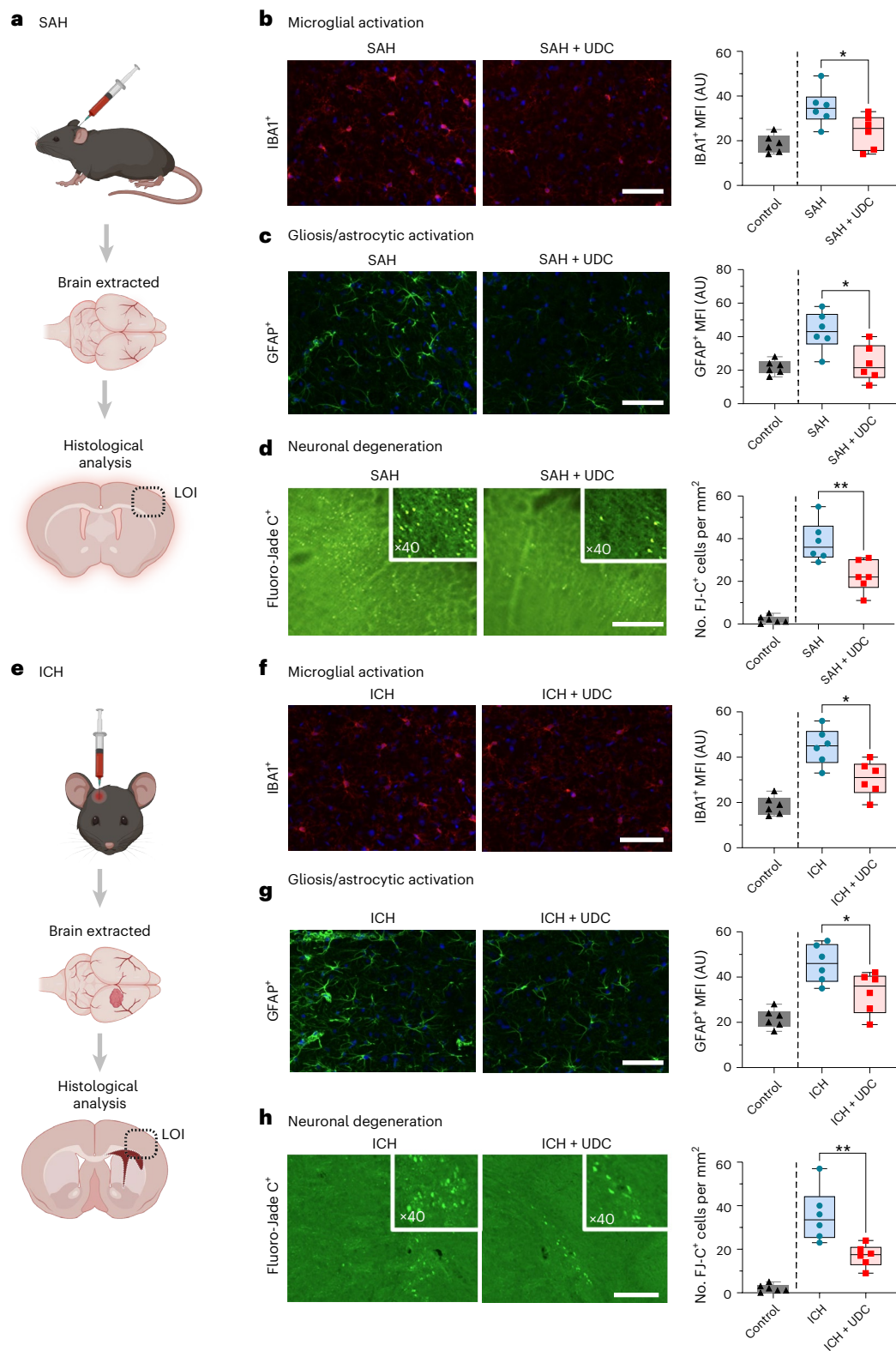


Fig. 2 | UDC reduces neuroinflammation and neurocytotoxicity after hemorrhagic brain injury. **a**, Schematic of the SAH model with the location of interest (LOI) sampled for histological analyses. **b,c**, Immunohistological assessment of SAH brains for microglial activation (IBA1, purple; **b**) and gliosis/astrocytic activation (GFAP, green; **c**) revealed a decrease in both neuroinflammatory markers with UDC compared to sham (IBA1: $P = 0.0411$, GFAP: $P = 0.0173$); MFI, mean fluorescence intensity. **d**, Histological assessment of SAH brains for neuronal degeneration (Fluoro-Jade C (FJ-C; green) revealed a decrease with UDC compared to sham ($P = 0.0087$). **e**, Schematic of the ICH model with the LOI sampled for histological analyses. **f,g**, Immunohistological assessment

of ICH brains for microglial activation (**f**) and gliosis/astrocytic activation (**g**) revealed a decrease in both neuroinflammatory markers with UDC compared to sham (IBA1: $P = 0.0260$, GFAP: $P = 0.0455$). **h**, Histological assessment of ICH brains for neuronal degeneration revealed a decrease with UDC compared to sham ($P = 0.0043$). Data presented as box plots that show minima, maxima, interquartile range (box bounds) and median (black line). Healthy control data are presented for reference and are repeated from **b–d** to **f–h**. Data were analyzed by two-tailed Mann–Whitney U -test; * $P \leq 0.05$, ** $P \leq 0.01$; $n = 6$ for each group; AU, arbitrary units. Scale bars, 100 μ m.

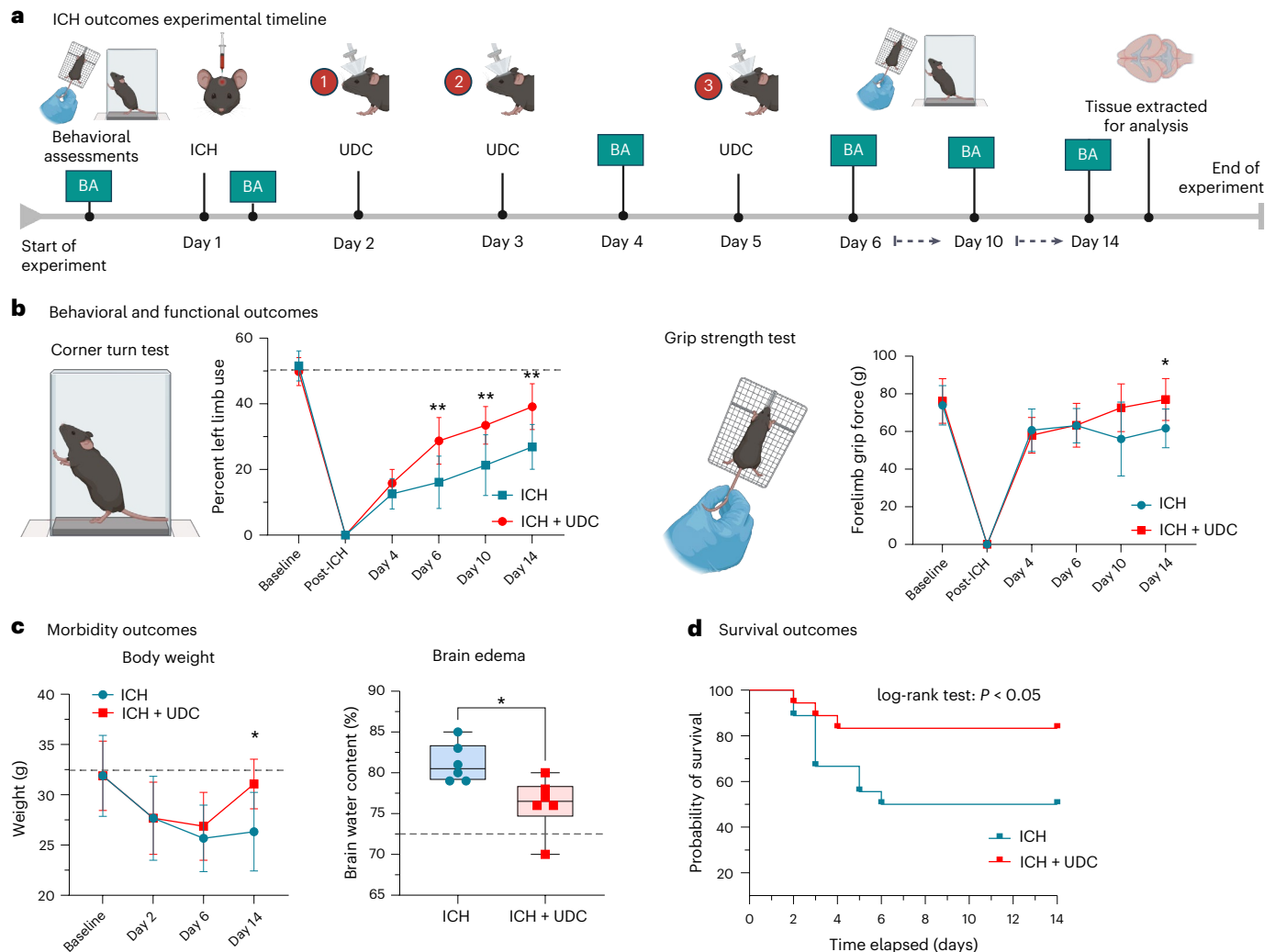


Fig. 3 | UDC improves behavioral outcomes and reduces morbidity and mortality after hemorrhagic brain injury. **a**, Timeline of ICH outcomes experiment. **b**, Behavioral/functional assessments (corner turn test and grip strength) revealed an improvement in functional outcomes as early as 6 days after hemorrhage with UDC compared to sham (corner turn test, day 6: $P = 0.0046$, day 10: $P = 0.0247$, day 14: $P = 0.0033$; grip strength test, day 10: $P = 0.0277$, day 14: $P = 0.0045$). **c**, Decreased morbidity outcomes (body weight and brain water content, edema) with UDC compared to sham (body weight, day 14: $P = 0.0035$; brain water content $P = 0.0108$). **d**, Increased survival with UDC intervention compared to sham; $\chi^2(2) = 4.08$, $P = 0.0432$. Time series data

points are presented as mean $\pm$ s.d., and two group comparisons presented as box plots show minima, maxima, interquartile range (box bounds) and median (black line); healthy control baseline levels are presented for reference (dashed lines, gray). Data in **b** and **c** were analyzed by Mann–Whitney U -tests with multiple comparisons controlled using the false discovery rate (FDR) method (two-stage step-up, Benjamini, Krieger and Yekutieli). Data in **c** were analyzed by Kruskal–Wallis H -test followed by a Dunn’s multiple comparisons test (left) or two-tailed Mann–Whitney U -test (right). Data in **d** were analyzed by log-rank (Mantel–Cox) test; $*P \leq 0.05$, $**P \leq 0.01$; $n = 18$ for each group; $n = 6$ for brain edema endpoint.

astrocytes that is strongly upregulated in reactive/activated astrocytes) was also reduced in UDC-treated animals in both models (mean fluorescence intensity $\pm$ s.d.; healthy control: 21.7 ± 3.85 ; SAH + sham: 43.33 ± 11.52 versus SAH + UDC: 24 ± 10.77 , ICH + sham: 46 ± 8.39 versus ICH + UDC: 33.17 ± 9.06 , $P \leq 0.05$; Fig. 2c,g). These results indicate that UDC effectively attenuates neuroinflammatory responses after hemorrhagic brain injury. Furthermore, histological assessment of neuronal degeneration using Fluoro-Jade C staining (a fluorescent dye that selectively labels degenerating neurons regardless of the mechanism of cell death) revealed a reduction in the number of degenerating neurons in UDC-treated animals compared to sham (mean $\pm$ s.d.; healthy control: 2.0 ± 1.63 ; SAH + sham: 38.5 ± 9.54 versus SAH + UDC: 22.5 ± 7.40 , ICH + sham: 35.5 ± 12.24 versus ICH + UDC: 17 ± 5.14 , $P \leq 0.01$; Fig. 2d,h). This reduction in neurotoxicity indicates the protective effects of noninvasive early blood clearance after hemorrhage in reducing neuronal damage.

Improved behavioral outcomes and survival with UDC treatment

Next, we assessed functional recovery and survival as indicators of UDC efficacy. Our pilot studies indicated that the SAH model did not produce a sustained behavioral deficit. Therefore, we conducted behavioral, morbidity and mortality outcomes analyses using the ICH model (Fig. 3a). UDC treatment improved both the assayed behavioral and functional outcomes (corner turn test and grip strength) as early as 6 days after hemorrhage, with sustained improvements observed through 14 days (mean $\pm$ s.d.; from post-ICH to day 14: corner turn test in percent left limb use: healthy control baseline: 50.7 ± 5.33 ; ICH + sham: 26.89 ± 6.83 versus ICH + UDC: 39.13 ± 7.09 ; grip strength in force per g: healthy control baseline: 89.8 ± 7.91 ; ICH + sham: 61.67 ± 10.27 versus ICH + UDC: 77 ± 11.51 , $P \leq 0.01$ compared to sham; Fig. 3b).

Morbidity outcomes, including body weight and brain water content (indicative of edema), were also assessed. UDC-treated mice

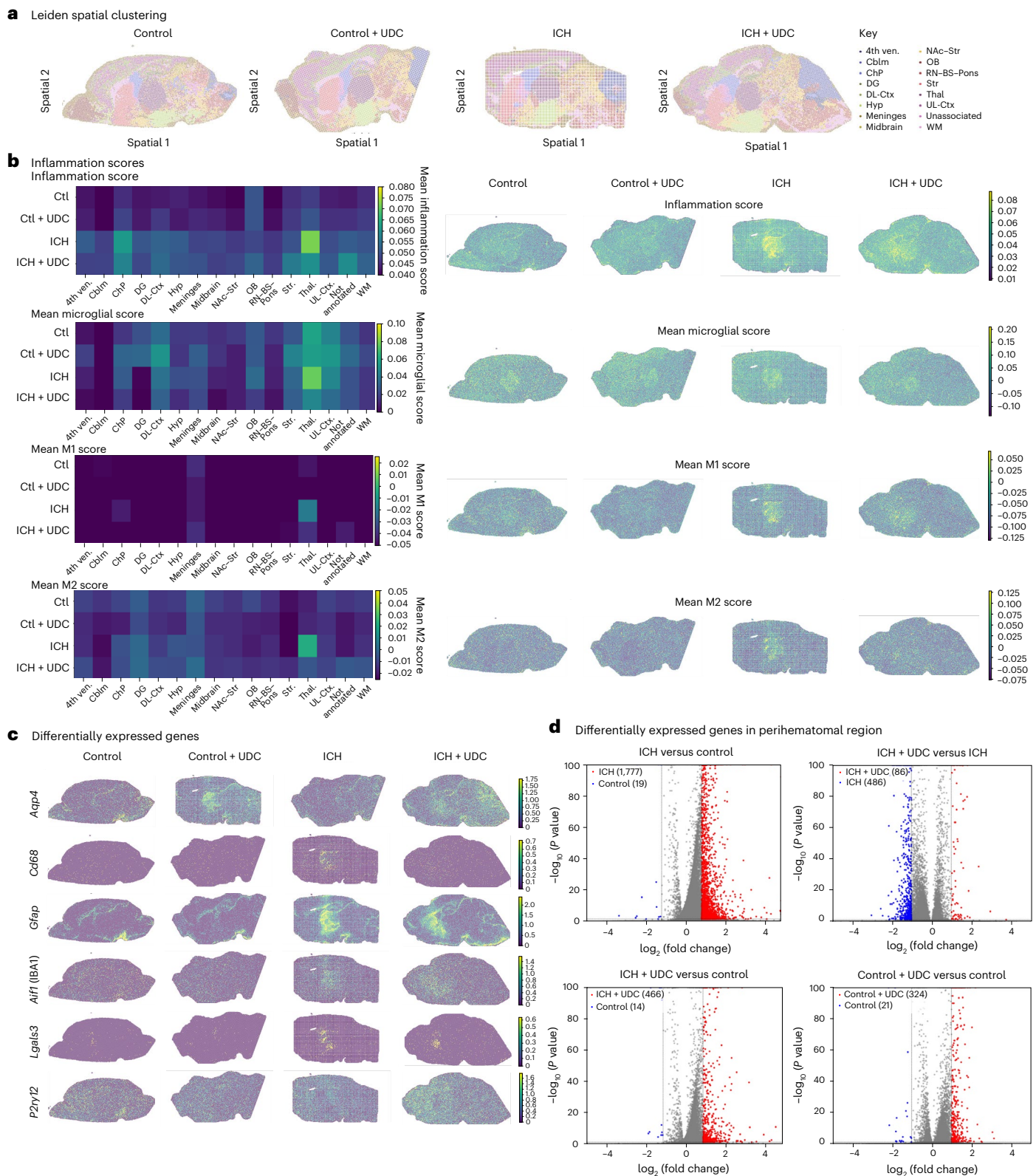


Fig. 4 | Spatial transcriptomic analysis demonstrates UDC-mediated changes in inflammatory markers after hemorrhagic stroke. a, Brain region-level annotations of Leiden-derived spatial clusters of stereotranscriptomic data at a bin size of 100. Bins are colored by annotation; 4th ven., fourth ventricle; Cblm, cerebellum; ChP, choroid plexus; DG, dentate gyrus; DL-Ctx, deep-layer cortex; Hyp, hypothalamus; NAc-Str, nucleus accumbens-striatum; OB, olfactory bulb; RN-BS-Pons, reticular nucleus/brain stem/pons; Str, striatum; Thal, thalamus; UL-Ctx, upper-layer cortex; WM, white matter. The striatum and thalamus regions together constitute the perihematomal regions of the ICH groups. **b**, Gene group

scoring across different brain regions and conditions (left: heat map; right: spatial distribution; inflammation score (GO:0006954) and microglia score (GO:1903978)); 'disease-associated' and 'homeostatic' scores are from manually curated lists (see Methods); Ctl, control. **c**, Key markers of glymphatics (*Aqp4*), glia (*Gfap*), microglia (*Aif1*, *P2yr12*) and inflammation (*Cd68*, *Lgals3*). **d**, Volcano plots of all genes across four different comparisons for data derived from the striatum and thalamus, which for ICH animals constitute the perihematomal region. The numbers of differentially expressed genes are shown in parentheses, determined as $|\log_2(\text{fold change})| > 1$; $q < 0.05$.

exhibited lower brain edema and recovered their body weight faster than sham-treated controls (mean $\pm$ s.d.; brain edema, as percent brain water content: healthy control baseline: 72.6 ± 3.40 ; ICH + sham: 81.17 ± 2.40 versus ICH + UDC: 76.17 ± 3.37 ; body weight in grams at 14 days: healthy control baseline: 32.4 ± 3.10 ; ICH + sham: 26.33 ± 3.91 versus ICH + UDC: 31.07 ± 2.54 , $P \leq 0.05$; Fig. 3c). Notably, UDC treatment increased survival rates compared to sham (83.3% versus 50% at 14 days, $P < 0.05$; Fig. 3d). These disparate survival rates between the sham and UDC groups support the positive behavioral effects (Fig. 3a,b), which were observed despite a survivorship bias in these data that otherwise serves to lessen the observable behavioral effect sizes. These findings demonstrate that UDC not only enhances functional recovery but also reduces morbidity and mortality after hemorrhagic brain injury.

To provide a benchmark comparison, we tested a pharmacologic agonist of mechanosensitive channels (Yoda-1) using prior protocols that induced CSF clearance²². In the ICH model, these protocols proved detrimental, with decreased survival of treated mice compared to sham (Extended Data Fig. 3a,b). Mice that survived to the endpoint showed no decrease of hematoma volume, indicating reduced clearance efficacy compared to UDC (Extended Data Fig. 3c). Thus, UDC is safer and more efficacious than a benchmark pharmacologic therapy in this setting.

UDC induces homeostatic neuroimmunologic changes and cellular glymphatic processes

To investigate the potential cellular and molecular effects induced by UDC, we applied a spatial transcriptomic analysis to brain tissue from both healthy control and ICH model mice treated with sham or UDC (Fig. 4a). Consistent with prior literature and our assessment of activated microglia and astrocyte states (Fig. 2), the ICH model induced prominent upregulation of neuroinflammation and microglial activation (Fig. 4b). In the ICH model, UDC reduced both inflammation and microglial scores compared to sham, particularly in the perihematomal region, with a similar magnitude of transcriptional change of both disease-associated and homeostatic microglial states (Fig. 4b). Investigating specific factors of interest, we observed substantial transcriptional upregulation of *Aqp4*, *Gfap*, *Cd68*, *Aif1* (IBA1), *Lgals3*, *P2ry6* and *P2ry14* in the ICH model, particularly in the perihematomal region. With UDC treatment of the ICH model, the neuroinflammatory genes *Cd68* and *Lgals3* showed relative downregulation in the perihematomal region (Fig. 4c), consistent with our prior analysis of reduced microglial activation with UDC (Fig. 2). By contrast, *P2ry12*, a homeostatic microglial marker, showed a trend toward increased expression with UDC treatment of ICH. For *Aqp4*, although transcript levels were not altered by UDC in the ICH model, we note that AQP4 is strongly expressed in astrocytic endfeet at the gliovascular interface, and changes in its protein localization or polarization may occur independently of transcriptional changes. At the transcriptional level, no definite change of *Aqp4* or *Gfap* was noted with UDC in the ICH model, in potential contrast to the observed changes in expression of GFAP protein (Fig. 2c,g). No definite changes of these select markers were noted after UDC in healthy control animals. The largest activation in perihematomal gene expression was observed between ICH and control

(1,777 genes upregulated and 19 genes downregulated in ICH). UDC restored ICH perihematomal gene expression toward its baseline, with substantial gene downregulation in ICH + UDC versus ICH (486 genes downregulated in ICH + UDC) and only modest gene expression differences between ICH + UDC and healthy control animals (466 genes upregulated and 14 genes downregulated in ICH + UDC). Additionally, UDC induced only limited gene expression changes in control animals (324 upregulated and 21 downregulated in control + UDC versus control), supporting the interpretation that its effects are specific to injury-related pathways (Fig. 4d). Together, these data suggest that UDC corrects ICH-associated gene networks, without substantial or concerning off-target gene activation in control conditions.

To determine whether these transcriptional changes were reflected at the protein level, we studied an immunohistochemistry panel in the ICH model with and without UDC treatment (Fig. 5a). First, given our findings that UDC treatment of hemorrhagic brain injury reduced GFAP protein (Fig. 2c,g), but not *Gfap* mRNA (Fig. 4c), expression, we assessed the expression of the independent astrocytic marker S100 β via immunohistochemistry, which showed no difference after UDC. This indicates that UDC does not reduce astrocyte numbers and that the decrease in GFAP protein expression after UDC may reflect either a biological discrepancy between *Gfap* mRNA and GFAP protein or a per astrocyte decrease of activation state (Fig. 5b)^{36,37}. Next, following the spatial transcriptomics data suggesting changes in microglial state (Fig. 4), we assessed markers of different microglial states identified in the literature^{38,39}, which include disease-associated microglia and homeostatic microglia. Microglial ramification branch complexity is a marker of microglial state, with high branch complexity correlating to the homeostatic state^{40,41}. UDC increased microglial branch complexity (mean branch levels $\pm$ s.d.; ICH + sham: 3.68 ± 0.28 versus ICH + UDC: 5.23 ± 0.77 , $P \leq 0.05$; Fig. 5c). In agreement with this result and the spatial transcriptomics analysis, CD68 and GAL-3, which are markers of microglial activation^{42,43}, were both decreased with UDC (mean number of cells $\pm$ s.d.; CD68: healthy control: 0.0 ± 0 ; ICH + sham: 401.5 ± 200.17 versus ICH + UDC: 45.63 ± 29.49 , GAL-3: healthy control: 0.0 ± 0 ; ICH + sham: 727.5 ± 66.75 versus ICH + UDC: 257 ± 131.97 , $P \leq 0.05$; Fig. 5d). Likewise, P2RY12, a cell surface marker of homeostatic microglia necessary for the clearing of cellular debris⁴⁴, increased with UDC (mean number of cells $\pm$ s.d.; healthy control: 25.5 ± 0.32 ; ICH + sham: 12.05 ± 3.84 versus ICH + UDC: 21.18 ± 0.95 , $P \leq 0.05$; Fig. 5e). Additionally, we observed a notable increase in the colocalization of red blood cell markers (TER-119⁺) with microglia (IBA1⁺) in the perihematoma region, suggesting elevated microglial phagocytosis of erythrocytes with UDC treatment of ICH (mean number of phagocytosed cells $\pm$ s.d.; ICH: 1.5 ± 0.58 versus ICH + UDC: 9.75 ± 2.5 , $P \leq 0.05$; Fig. 5f and Supplementary Video 1). Finally, in contrast to the overall transcriptomic upregulation of *Aqp4* in ICH (Fig. 4c), we observed a nonstatistically significant trend toward decreased astrocytic endfeet perivascular polarization of AQP4 after ICH (Fig. 5g). AQP4 endfeet perivascular polarization, a key requirement of glymphatic upregulation, was restored in ICH models with UDC treatment (mean fluorescence intensity $\pm$ s.d.; healthy control: 37.15 ± 3.04 ; ICH + sham: 21.03 ± 7.53 versus ICH + UDC: 50.8 ± 11.80 , $P \leq 0.05$). Considering the

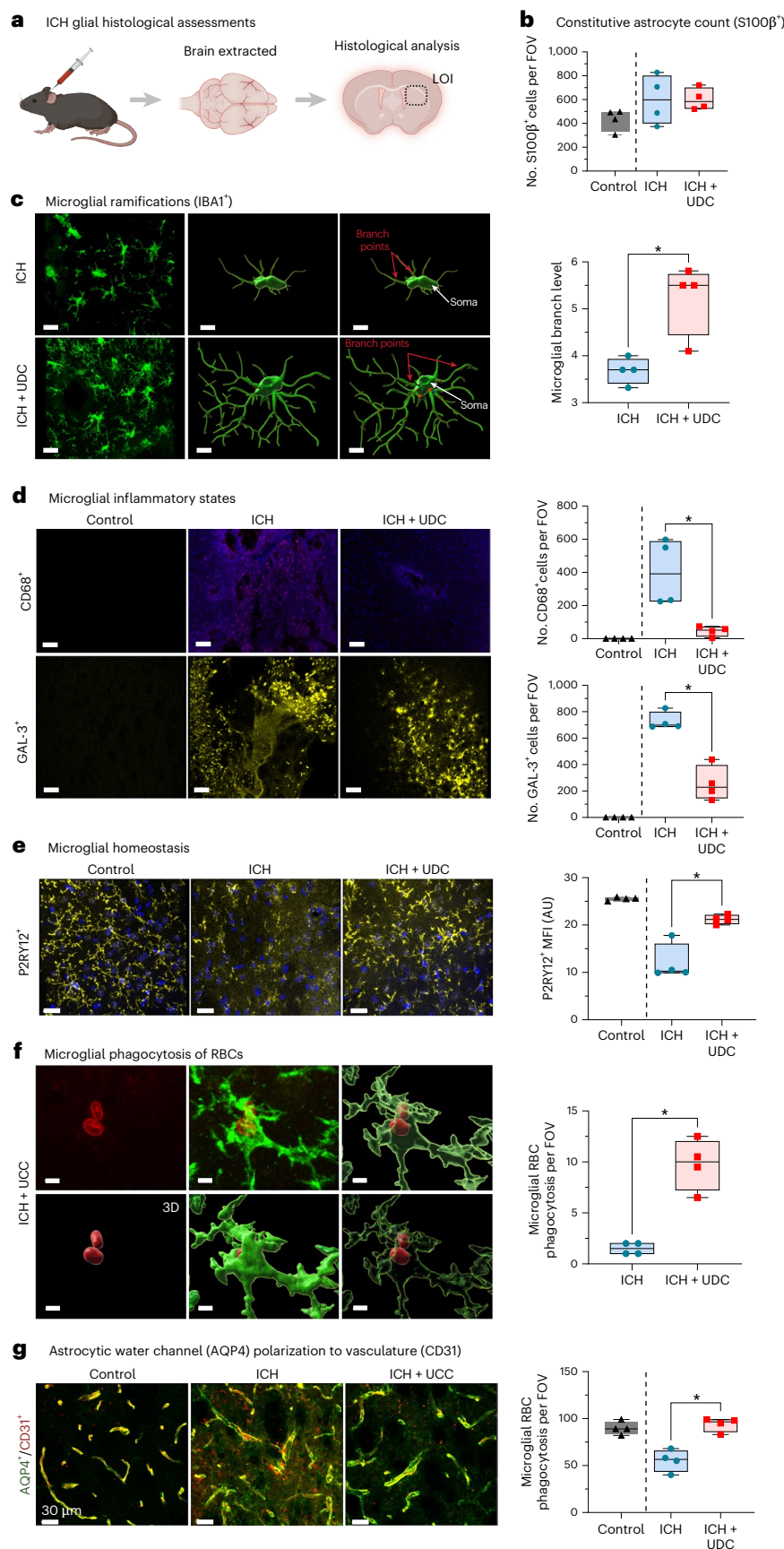
Fig. 5 | UDC promotes blood clearance by modulating neuroinflammatory and glymphatic states. **a**, ICH model with the LOI sampled for histological analyses. **b**, Immunohistological assessment of ICH brains for constitutive astrocytic counts (S100 β) revealed no significant difference with ICH + UDC compared to ICH + sham ($P = 0.1455$); FOV, field of view. **c**, Immunohistological assessment with three-dimensional (3D) reconstruction of microglia revealed increased branching, a hallmark of the homeostatic microglial state, with ICH + UDC compared to ICH + sham ($P = 0.0286$). Scale bars, 5 μ m. **d**, Immunohistological assessment of microglial inflammatory states (CD68⁺, red; GAL-3⁺, yellow) revealed decreased proinflammatory disease-associated microglial markers with ICH + UDC compared to ICH + sham (CD68: $P = 0.0286$; GAL-3: $P = 0.0286$). CD68⁺ scale bars, 50 μ m; GAL-3⁺ scale bars, 100 μ m. **e**, Immunohistological assessment

of purinergic receptors (P2RY12⁺, yellow) that reflect a homeostatic microglial state revealed an increase with ICH + UDC with a trend toward the healthy controls compared to ICH + sham ($P = 0.0286$). Scale bars, 20 μ m. **f**, Immunohistological assessment with 3D reconstruction of microglia (IBA1⁺, green) revealed increased phagocytosis of red blood cells (TER-119⁺, red) with ICH + UDC compared to ICH + sham ($P = 0.0286$). Scale bars, 5 μ m. **g**, Immunohistological assessment of astrocytic endfeet water channel localization (AQP4⁺, green) revealed increased polarization to the vasculature with ICH + UDC compared to ICH + sham ($P = 0.0286$). Data presented as box plots that show minima, maxima, interquartile range (box bounds) and median (black line). Data were analyzed by two-tailed Mann–Whitney *U*-test; * $P \leq 0.05$; $n = 4$ for each group.

totality of transcriptomic and immunohistochemistry data, we conclude that UDC decreases neuroinflammation via shifting microglia to the homeostatic phenotype and upregulates glymphatic function by increasing perivascular polarization of AQP4 in astrocytic endfeet.

Ultrasound-mediated debris clearance acts through molecular mechanotransduction pathways

To further investigate the underlying mechanisms of ultrasound-mediated clearance, we considered potential thermal effects



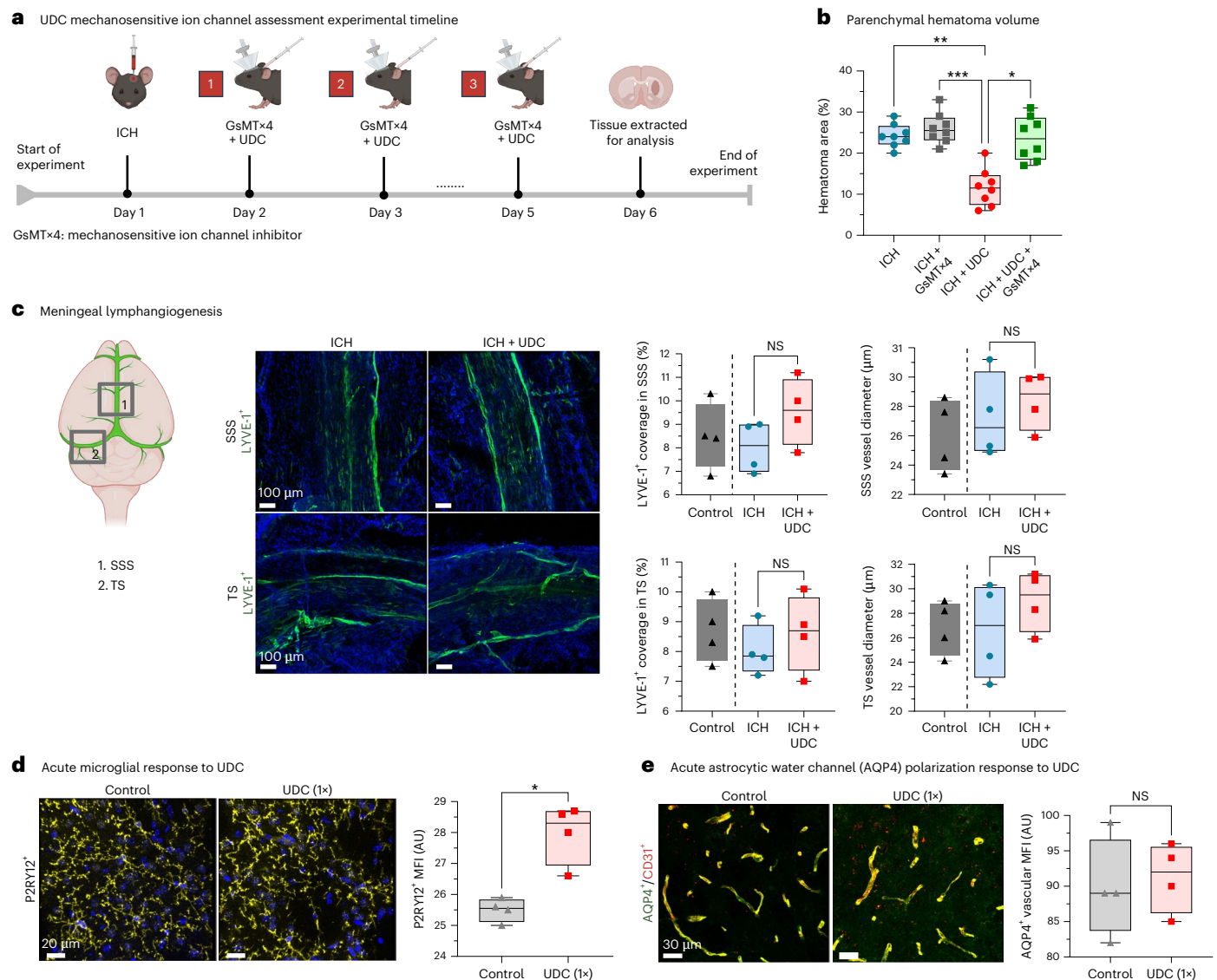


Fig. 6 | UDC is reversed by a mechanosensitive ion channel blocker. a, Timeline of ICH experiment with UDC and a mechanosensitive ion channel blocker (GsMTx4). **b**, Administration of GsMTx4 reduced the blood product clearance of UDC, as assessed by hematoma volumes across treatment groups. **c**, Immunohistological analysis of meningeal lymphangiogenesis (LYVE-1⁺) revealed no significant difference between control, ICH + sham and ICH + UDC during this timeline of injury and treatment; SSS, superior sagittal sinus; TS, transverse sinus; NS, not significant. **d**, In healthy, uninjured mice, immunohistological analysis of purinergic receptors (P2RY12⁺, yellow) that

reflect a homeostatic microglial state revealed an acute increase with only one application of UDC compared to control ($P = 0.0286$). **e**, In healthy, uninjured mice, immunohistological analysis of AQP4 (green) at astrocytic endfeet revealed no difference in polarization with one application of UDC compared to control ($P = 0.6286$). Data presented as box plots that show minima, maxima, interquartile range (box bounds) and median (black line); healthy control data are presented for reference. Data were analyzed by Kruskal–Wallis test with a post hoc Dunn's multiple comparisons test; * $P \leq 0.05$, ** $P \leq 0.01$, *** $P \leq 0.001$; $n = 8$ mice for each group in Fig. 6b and $n = 4$ mice for each group in Fig. 6c–e.

(Extended Data Fig. 2). At the intensity, duty cycle and ultrasound frequency used here, computational simulations (via k-Wave⁴⁵) indicated no substantive parenchymal heating, with heat accumulation being minimal and preferentially in the skull, not the parenchyma. We confirmed this experimentally by measuring brain temperature before and after protocol application with a thermocouple placed intracranially, immediately under the skull. We noted a cooling trend over the experiment, presumably due to thermal conduction into the room temperature gel and water used for ultrasound coupling in contact with the dorsal scalp (peak intracranial temperature change: -0.5 ± 0.2 °C; Extended Data Fig. 2).

Given this result, we considered possible mechanical mechanisms such as direct induction of convection or oscillation of the intracranial fluid medium that promotes exchange of the CSF and interstitial fluid (as in Extended Data Fig. 1)²⁵, with or without an accompanying

mechanical change of the parenchyma that promotes downstream CSF circulation⁴⁶. However, our prior results demonstrated that the response to UDC extends for over 1 h after ultrasound exposure²⁵, implying a more prolonged mechanism. Because the ultrasound intensity of our protocol would be sufficient to activate mechanosensitive channels^{16,17}, which can induce meningeal lymphatic vessel sprouting and increased CSF egress^{21,22}, we investigated the effects of UDC on intracranial mechanosensitive channels. Using the ICH model, we cisternally administered a broad-spectrum mechanosensitive ion channel blocker (GsMTx4, a selective blocker of Piezo and TRP family channels; Fig. 6a) and observed elimination of the blood cell clearance efficacy of UDC, as evidenced by increased hematoma volumes in the UDC + GsMTx4 treatment group versus UDC treatment alone (mean hematoma area percent $\pm$ s.d.; ICH + UDC: 11.63 ± 4.59 versus ICH + UDC + GsMTx4: 23.63 ± 5.50) and no difference between the

sham + GsMTx4 and UDC + GsMTx4 groups (Fig. 6b). The elimination of detectable clearance with GsMTx4 blockade (Hedge's *g*, corrected for $N \leq 50 = 2.96$ for sham versus UDC alone compared to 0.46 for sham + GsMTx4 versus UDC + GsMTx4) indicates that mechanosensitive ion channels mostly mediate the clearance effects of UDC. We observed no change in meningeal lymphatic vessel coverage or caliber in the ICH model or in response to UDC treatment (Fig. 6c), potentially reflecting the short, days-long timeline of both this disease model and the intervention. By contrast, we observed that UDC, when applied only once to healthy control brains, acutely increased P2RY12⁺ microglia, a marker of the homeostatic state⁴⁴ (mean fluorescence intensity $\pm$ s.d.; healthy controls: 25.5 ± 0.37 versus healthy controls + UDC (one time): 27.98 ± 0.97 , $P \leq 0.05$; Fig. 6d), supporting the role of mechanosensitive channels in UDC efficacy. Notably, AQP4 expression did not change in healthy control brains after only one application of UDC (Fig. 6e), suggesting that UDC acts indirectly to drive the observed glymphatic enhancement in ICH (Fig. 5g). Instead, AQP4 polarization with UDC treatment of ICH may be secondarily induced through more direct UDC-driven reductions in neuroinflammation, with secondary disinhibition of AQP4 polarization, suggesting a distinction between the direct immunomodulatory effects of UDC and its relatively indirect glymphatic effects of AQP4 polarization.

Discussion

We have demonstrated that our UDC protocol facilitates the clearance of cellular debris from the mouse brain, reduces neuroinflammation and neurocytotoxicity and improves functional outcomes and survival after hemorrhagic brain injury (Figs. 1–3). UDC was safer and more efficacious than a pharmacologic benchmark for this indication (Extended Data Fig. 3). Our mechanistic studies suggest (Extended Data Fig. 4) that the clearance effects of UDC depend primarily on activation of mechanosensitive channels (Fig. 6a,b) and increased intracranial fluid compartmental intermixing (Extended Data Fig. 1)²⁵. Activation of mechanosensitive channels shifts the microglial phenotype to the homeostatic state (Figs. 5c–e and 6d), promoting debris clearance in both the CSF and interstitial compartments (Figs. 1 and 5f and Supplementary Video 1). This shifting neuroinflammatory milieu promotes increased astrocytic endfeet perivascular polarization (Fig. 5g) by disinhibiting polarization in the setting of increased aquaporin expression after hemorrhage (Fig. 4). Our data suggest a model (Extended Data Fig. 4) in which decreased neuroinflammation, increased fluid compartment intermixing and increased glymphatic mediators, like aquaporin astrocytic endfeet perivascular polarization, combine to clear debris from the central nervous system to the lymphatic system of the body (Fig. 1), which in turn lowers morbidity and mortality in hemorrhagic stroke models (Figs. 2 and 3).

It should be emphasized that this model is still hypothetical, and several aspects require further investigation and validation in future studies. For instance, the mechanosensitive channel antagonist GsMTx4, which blocks UDC efficacy (Fig. 6a,b), limits mechanotransduction at the level of the lipid bilayer and therefore antagonizes the broad general class of mechanosensitive channels, including most, if not all, of the Piezo and TRP families^{47,48}, making it unclear which specific channels (Piezo1, Piezo2, TRPA1, TRPC6 and so on) most contribute to UDC efficacy. It is also worth noting that we did not observe changes in lymphatic vessel coverage or morphology with UDC (Fig. 6c), although mechanosensitive channel activation can stimulate lymphatic flow without lymphatic morphological change²¹. Additionally, as mechanical stimulation of the face and neck can increase lymphatic drainage of the central nervous system⁴⁹ and UDC is a form of mechanical stimulation, this mechanism may partly contribute to UDC efficacy.

We designed our UDC protocol to be compatible with clinical translation. The 250-kHz ultrasound frequency we used is in the range of commercial clinical transcranial FUS systems on the market

worldwide⁵⁰. This ultrasound frequency has relatively low attenuation and aberration through the human scalp and skull^{28,29}. Additionally, the target in situ peak negative pressure we used corresponds to a mechanical index (a metric indicating the risk of mechanical injury induced by ultrasound) of 0.9, which is well within Food and Drug Administration guidelines for safe application⁵¹ and similar to that of recent studies of ultrasound neuromodulation^{51,52}. This suggests that clinical trials of UDC should receive a nonsignificant risk designation from local institutional review boards.

If debris clearance by UDC is validated in clinical trials, it would not incur the risks of invasive surgical approaches nor the systemic toxicities of pharmacologic treatments and would not require coadministration of exogenous agents like nanoparticles or microbubbles⁵³, simplifying clinical application and increasing its safe operating regime. As its mechanism does not require a particular pattern of brain activity (that is, it is effective in anesthetized animals), it would be applicable in humans who are in a range of cognitive states, including awake, asleep, engaged in a task or obtunded, in the immediate postinjury recovery period. Furthermore, it would provide a method to drive CSF circulation and egress complementary to pharmacologic approaches and approaches that activate particular neural activity patterns that are correlated with this circulation^{54,55}. Accordingly, it would offer an alternate method for understanding and intervening in glymphatic CSF circulation and egress in the variety of disorders and normal states in which it has been implicated, including sleep⁵⁶, neurodegeneration⁵⁷, traumatic brain injury⁵⁸ and mental health conditions⁵⁹.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41587-025-02866-8>.

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